QUALITY ASSURANCE TESTING DOCUMENTATION

(QATD)

UMHackathon 2026

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Document Control

Field	Detail
System Under Test (SUT)	*UMHackathon 2026 - Team **Teapot***
Team Repo URL	https://github.com/Nikolrize/teapot-umhackathon-backend *https://github.com/Nikolrize/teapot-umhackathon-frontend*
Project Board URL	-
Live Deployment URL (optional for preliminary round)	*Backend: https://teapot-umhackathon-backend.onrender.com/docs#/* Frontend: https://teapot-umhackathon-frontend.vercel.app/

Objective:

*The primary objective is to ensure that **Teapot** can handle*

PRELIMINARY ROUND (Test Strategy & Planning)

1. Scope & Requirements Traceability

*This section aligns the testing connected back to specific user requirements via **Requirement Traceability Matrix".** So, it ensures every test has been conducted to prevent bugs/failure in products for that specific requirements and unplanned feature creep.*

1.1 In-Scope Core Features

- **Multi-Agent Ecosystem**
    - **Default Agents:** Provision of 7 specialized pre-configured agents to assist with diverse business operations.
    - **Custom Agent Builder:** Administrative capability to create new agents using

specific LLM backends, including **Z.ai, Gemini, GPT models**, etc.

- **Agent Configuration:** Admin can control for setting conversation starters, "pass requirements," and behavior logic for each agent.

- Advanced Dashboard Systems
  - **User Dashboard:** A personalized workspace allowing users to pin and organize specific agent responses for quick reference and business use.
  - **Admin Analytics Dashboard:** High-level tracking of business metrics including Total Leads, New Leads, Converted Leads, Lost Leads, and Conversion Rate.
- Resource & Access Management
  - **Token Allocation:** Admin control over token limits for specific AI models and individual user accounts to manage costs and usage.
  - **User Management:** Backend systems for account registration, status management, and profile (avatar) customization.

1.2 Out-of-Scope

- Communication Tools
  - **Content Sharing:** Ability for users to share specific chat logs or agent interactions with other users within the Teapot system

- Profile Management:
  - Edit profile info, change Avatar

2. Risk Assessment & Mitigation Strategy

*[e.g. Quality Assurance risks are anticipatory. So, identify the technical risk associated with you application requirements, architecture. Now, the risk needs to be evaluated using the 5*5 Risk assessment Matrix. So, Risk Score = Likelihood * Severity.]*

Technical Risk	Likelihood (1–5)	Severity (1–5)	Risk Score (L×S)	Mitigation Strategy	Testing Approach
User able to enter the ai model that is	*5*	*3*	*15*	*Make the model selection as drop down instead of text*	*Use mocked user account to see is the user still able to*

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Technical Risk	Likelihood (1–5)	Severity (1–5)	Risk Score (L×S)	Mitigation Strategy	Testing Approach
not available in the backend				*field*	*access to the model*

Risk Assessment Scoring Criteria:

3. Test Environment & Execution Strategy

Explain the testing environment. That means where the testing of your application will take place, how you are going to handle test data and the rules/threshold that you are incorporating to govern your testing phases.

- **Unit Test**
  - *Scope: Individual backend logic including Token Calculation, Input Validation (e.g., ensuring prompt length isn't exceeded), Avatar Metadata Check (validating file extensions before upload) and more.*
  - *Execution: Tests are written using the PyTest framework. Execution occurs locally during development and automatically via GitHub Actions during the CI pipeline.*
  - *Isolation: External dependencies such as the PostgreSQL database and Cloudinary API are mocked using unittest.mock to ensure we are testing the Python logic in isolation.*
  - *Pass Condition: 100% pass rate for "Happy Path" (valid inputs) and "Negative Cases" (e.g., ensuring an error is thrown if a user exceeds their assigned token limit).*
- **Integration Test**
  - *Scope: Agent-to-LLM Workflow and Frontend-to-Dashboard Sync.*
  - *Execution: Performed after unit tests pass. This involves testing the full cycle from the User sending a message to the Admin seeing the updated "Converted Leads" on their dashboard*
  - *Workflow: Real API calls are made to the database and internal service endpoints. For the LLM (Z.ai, Gemini, GPT), a limited "Test API Key" is used to ensure the connection and parsing logic are functional.*
  - *Pass Condition: Successful end-to-end data flow (e.g., a pinned reply in the User Dashboard correctly displays the Cloudinary-hosted image without broken links).*
- **Test Environment (CI/CD Practice):**
  - *Local Testing: Developers run the Teapot system on localhost:8000*
  - *Staging/CI: GitHub Actions is utilized to trigger automated suites on every push.*
  - *CI/CD Automated Pipeline: GitHub Actions performs linting and basic connectivity tests on feature branches. A full PyTest suite is required to pass before a Pull Request (PR) can be merged into the main branch.*

- **Regression Testing & Pass/Fail Rules:**
  - *Execution Phase: A full regression suite is executed across all 7 default agents and the Admin configuration settings every time a merge to main occurs.*
  - *Pass/Fail Condition: A test is "Pass" only if the actual outcome aligns 100% with expected results. As per project standards, a Minimum 90% Pass Rate across the entire*

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suite is required to allow a production deployment. Any failure is documented in the Defect Log.

- *Continuation Rule: Testing follows a "Gate" system; core authentication and database connection tests must hit a 100% threshold before the system proceeds to test complex AI Agent behaviors.*

- **Test Data Strategy:**
  - *Manual: Created 10 Mock User Accounts with varying roles (e.g., 2 Admins, 8 Client Users) to test dashboard visibility and permission restrictions.*
  - *Automated: SQL seed scripts can be used an input for the automated uses of dummy data*

- **Passing Rate Threshold:**
  - *A minimum of 86.2% of all test cases that are executed must pass. For the critical tests, all the tests (100%) need to be passed and there is no exception for that. Because critical failure costs huge in production*

4. CI/CD Release Thresholds & Automation Gates

This section is mainly for defining metrics & threshold for success criteria. So, if a certain threshold is not met, the pipeline blocks the forward transition.

4.1 Integration Thresholds (Merging to Main)

The primary needs of this threshold to be crucial is because the main branch must be a stable source of truth. So, it needs to be achieved through continuous integration checks.

Checks	***Requirements***	***Project***	***Pass/Failed***
Automatic Build	***Zero Build Error***	***0 Error***	***Passed***
Unit Tests	***100% Passing Rate***	***100%***	***Passed***
Code Quality	***Zero Linting Error***	***100%***	***Passed***
Test Coverage	***Minimum 81.2%***	***100%***	***Passed***

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4.2 Deployment Thresholds (Pushing to Production)

Checks	***Requirements***	***Project***	***Pass/Failed***
Regression Test	***Minimum 90%***	***98%***	***Passed***
AI Output Pass Rate	***Minimum of 80% of documented prompt***	***95%***	***Passed***
Critical Bugs	***Zero P0/01 Bugs***	***0***	***Passed***
API Performance	***Response Time < 800ms***	***≈500ms***	***Passed***
Security	***API keys are not exposed***	***Clean***	***Passed***

5. Test Case Specifications (Drafts)

This section is for you to draft your test cases. The test cases MUST include at least:

- ***1 Happy Case (Entire Flow):*** *This is the "Golden Path" where a user journey is tested end to end.*
- ***1 Specific Case (Negative/Edge Test):*** *This case should portray a scenario which triggers error in the system and the system handles it with proper handling techniques.*
- **2 Non-Functional (NFR) Tests (Performance/Load)*****:*** *This test consists of non-feature-based tests but architectural tests.*

Test Case ID	**Test Type & Mapped Feature**	**Test Description**	**Test Steps**	**Expected Result**	**Actual Result**
TC-01	***Happy Case (Entire Flow):*** *Create Project and chat with the agent*	*Verify whether a user can start a project and choose an agent to share about the business idea and share the discussion with other users and pin the reply of the ai agent to the*	*1. Open Website, login with credentials.* *2. choose and agent.* *3. start to chat with the ai agents.*	*The reply is successfully generated by the ai agent, the uploaded file is successfully read by the ai agent. The discussion is successfully*	*Success. The user can login into the system and start a project and start a disscussion with the ai agent and it can pin the reply and*

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Test Case ID	Test Type & Mapped Feature	Test Description	Test Steps	Expected Result	Actual Result
		dashboard. So, all the modules are communicating with each other correctly.	*4. upload files like pdf or image to the ai to analyses.* *5. share the discussion to the chat with others user.* *6.pin the reply of the ai agent to the dashboard.*	*share to others users. Last is the reply of the ai agent is successfully pin on the dashboard.*	*share to others user.*
TC-02	***Specific Case (Negative):*** *invalid avatar file type*	*Verify the system does handle error quite optimally and block access when an invalid file type in entered to the system.*	*1. Go to the profile* *2, go to the edit profile* *3 go to change avatar and select file* *4. select random file type like .txt*	*The upload process is block and display only certain file type is allow to upload to avatar*	*The file is successfully block and the error message is successfully displayed to the user* *Status: success*
TC-03	***NFR (Performance):*** *API Response Test*	*Payment API remains quite responsive (< 800ms) when concurrent users are using.*	*1. Open Postman, selecting the API Endpoints.* *2. 50 users GET Requests to that endpoint.* *3. Record the average response time.*	*Average response time < 800ms in addition 0% error rate.*	*Average Response time is received as 1328ms. 0% Error Rate.* *Status: Failed*

6. AI Output & Boundary Testing (Drafts)

This section is designed to validate that your GLM integration does produce acceptable output and handles any inputs that are abnormal gracefully.

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6.1. **Prompt/Response Test Pairs**

Document at least 3 Prompts/response test pairs. For each of them, do define the acceptance criteria - what a correct, acceptable or a failing response may look like for your use case.

Test ID	Agent	Prompt Input	Expected output	Actual Output	Status
001	sales-predictor agent	help me to predict the sales of my milk tea business base on the test business mode and budgets		As an experienced sales analyst, I can certainly help you lay the groundwork for predicting sales and identifying revenue opportunities for your milk tea business. However, the provided project context ("Test," "tester," "test business mode") is extremely generic. To provide a truly accurate and actionable prediction, I need more specific details about your business. Without historical	

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				sales data, specific operational details, or a clear definition of your "test business mode," any prediction would be purely speculative. Let's approach this by first outlining the critical information needed for a robust sales forecast, then discussing common sales drivers and revenue opportunities for a milk tea business, and finally, how your budget of up to $250,000 can be strategically utilized.	
002	--scenario-simulator agent	In the current world case of the fights and wars, how would my metal import and outport business be affected		The current global landscape, marked by "fights and wars," introduces significant volatility and	

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				risk for a metal import and export business. The specific impact will depend heavily on the geographic scope, intensity, and duration of these conflicts, as well as the specific metals you trade and your supply chain's resilience. Here's a simulation of best-case, worst-case, and most likely business scenarios for your metal import/export operation: ---	
003	--decision-recomendation agent	can you help me to determine how should the company increase revenue, by expanding to more domain, or further dive in current domain sales_data.csv CSV		Based on the provided sales data and your goal of increasing revenue, the best next strategic decision for your business is to **further dive into the	

				current domain.** Here's the strategic rationale and recommended actions: **Strategic Rationale:**	

6.2. Oversized/Larger Input Test

In this section, do define the maximum input size your system accepts. Do highlight here what happens when the limit is exceeded.

Fields	
Maximum Input Size	
Input used while testing	
Expected Behavior	
Actual Behavior	
Status	

6.3. Adversarial/Edge Prompt Test

In this section, test minimum of 1 with one prompt that is ambiguous, malformed - document how GLM does respond and how your system is going to handle it.

6.4. Hallucinating Handling

This section is defined for you to describe how your system does detect or mitigate hallucinating response.

Important Note:

Using Agile principle, these testing will take place iteratively over the period of development, not limited to at the end.

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